YUANJUN (JOHNNY) DAI

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SUMMARY

AI Infrastructure and Applied AI Engineer, **PhD candidate** at CWRU. Specializes in the security, optimization, and observability of ML/LLM training and inference infrastructure. 8 research papers (7 first-author) at IEEE, ACM, and Springer; **2+ years of systems software engineering** at Cisco and Broadcom; 2+ years of **cross-functional** product ownership at Midea and Xiaomi.

EDUCATION

Ph.D. Candidate in Computer Science (AI Infrastructure) 08/2019 – Present
Case Western Reserve University

M.S. in Electrical and Computer Engineering (Computer Architecture)
University of Pittsburgh

B.S. in Electrical Engineering
Northeastern University

SKILLS

Languages	Python, C/C++
ML Frameworks	PyTorch (DDP/FSDP), vLLM, DeepSpeed, Megatron-LM, NCCL, BytePS, scikit-learn
Parallelism Paradigms	Data Parallel (DDP/FSDP), Model Parallel (TP/PP), Pipeline Parallel (DeepSpeed), Hybrid Parallel
Infrastructure	Kubernetes, Docker, CI/CD Pipelines, ArgoCD, Slurm, HPC
Systems & APIs	FastAPI, REST API, gRPC
Observability	eBPF (Kernel Tracing), Prometheus, Grafana, DCGM Exporter, Intel RAPL
Cloud & HPC	Lambda Labs HPC, Ohio Supercomputer Center, NSF FABRIC, CloudLab
Networking	SDN (Software-Defined Networking), P4, NPU/ASIC (Broadcom BCM SDK)

RESEARCH & SYSTEMS ENGINEERING EXPERIENCE

Research Assistant, AI Infrastructure & ML Systems 08/2019 – Sep. 2026 (exp.)
Case Western Reserve University

Research focus: optimization, observability, and security of distributed ML/LLM training and inference infrastructure.

- **DYNAMIX – RL-Based Adaptive Batch Scheduling for Distributed ML** | Paper / Code
  Designed a PPO-based centralized RL scheduler monitoring real-time hardware signals (CPU/memory usage, network bandwidth, retransmission rate via custom eBPF probes) and statistical efficiency signals (loss convergence rate, generalization performance) to dynamically adapt batch size; co-scales learning rate proportionally (Linear Scaling Rule) to maintain training stability; evaluated on CNN architectures across CIFAR-100 and similar benchmarks using both Ring AllReduce (PyTorch DDP) and Parameter Server (BytePS) across **8–64 A100 GPUs**; achieves **46% end-to-end training time reduction** and improved final model accuracy over fixed batch size baseline.
- **PRISM – Priority-Based Selective Reliability for Distributed ML Training** | Paper (Under Review)
  Designed PRISM, a transport-layer library (NCCL/Gloo → UDP) hooking into PyTorch DDP's AllReduce at bucket-ready events to provide priority-based selective reliability without modifying training code; classifies gradients by magnitude into high-priority (full SACK-based retransmission) and low-priority (local compensation via cached values); supports kernel-bypass (LibVMA) and kernel-based (GSO/GRO) variants; achieves **8–15% throughput gain for CNNs and 15–21% for LLMs** over TCP under packet loss, matching NDP latency at 144-worker scale on commodity hardware.
- **Scalable LLM Serving Infrastructure on Kubernetes** | Code
  Independently designed and implemented end-to-end a K8s microservices platform (Gateway / Router / Worker) supporting vLLM inference with ArgoCD GitOps, Prometheus/Grafana observability, and CI/CD automation across **50+ Kubernetes deployments.**
- **Domain-Adapted LLMs for Clinical Risk Adjustment Coding** | Paper
  Designed an end-to-end chronic disease code classifier (115 HCC codes) from unstructured EHR notes on an extremely imbalanced multi-label dataset (MIMIC-V); built a semi-supervised labeling pipeline using **few-shot prompting with GPT-4o** (CMS definitions as context, domain expert verification) to construct training labels; replaced standard self-attention with **Fast Attention** ($O(Ld)$ vs $O(L^2)$) to support longer clinical sequences; applied **inverse-frequency per-label positive weighting** to counter severe class imbalance, and independently tuned a **decision threshold per label** for calibration and interpretability; fine-tuned **PubMedBERT via NVIDIA NeMo** and **Qwen-7B via Megatron-LM + DeepSpeed** on Lambda Labs A100 GPUs; achieves **macro-/ micro-F1 of 0.775/0.742** and macro-AUC of 0.9362.
- **Priority-Based eBPF Network Flow Telemetry for AI / DML Infrastructure** | Paper / Code
  Designed and implemented PSketch, the first in-kernel priority-aware sketch data structure using eBPF; high-priority flows tracked precisely via BPF_HASH priority table ($O(1)$ lookup), low-priority flows estimated via 3-layer Count-Min Sketch; voting-based hash

collision resolution and atomic operations for thread-safe in-kernel updates; achieves **96% Top-K accuracy, 96.4% retransmission recall**, and <1.1% throughput overhead at 10 Gbps on FABRIC testbed with CAIDA 2019 backbone traffic.

- **High-Resolution eBPF System Resource Profiling for DML Workloads** | Paper / Code
  Designed eHashPipe, a dual-pipeline eBPF observability framework for DML workloads — (1) a HashPipe-based memory profiler tracking Top-K memory-intensive processes via cascading slot eviction with ptr2size deallocation tracking, and (2) a per-thread CPU tracker hooking sched_switch tracepoint for thread-level CPU time attribution aggregated per PID; atomic operations for thread-safe in-kernel updates; delivers **10–14× finer temporal resolution** than top/htop, ~11ms latency, and >90% Top-K fidelity on commodity Linux servers.
- **Training-Time Side-Channel Attacks on Distributed ML Systems** | Paper / Code
  Developed deep understanding of DML computation and communication behavior in Parameter Server architecture to expose MLaaS training-phase side-channel vulnerability; (1) applied **K-means clustering** on push/pull network traffic to identify silent periods (forward pass computation without communication); (2) applied **Fourier transform** on Intel RAPL CPU and memory power signals for noise reduction, extracting layer-by-layer energy consumption patterns; (3) trained a **Decision Tree classifier** on energy features across Conv and FC layers under varying parameter sizes for layer-type classification; (4) built **polynomial regression models** on CPU energy profiles of activation functions for activation function classification; reconstructed complete target DNN architecture; achieves **>99% layer-type accuracy and <1.5% tensor-size error** across ZFNet, AlexNet, and VGG.
- **Scotch – Elastic SDN Overlay for Control-Plane Scalability under DDoS** | Paper / Code
  Designed Scotch, an Open vSwitch-based elastic overlay for SDN control-plane scalability under DDoS attacks — when physical switch OFA capacity is saturated, tunnels new flows to a vSwitch pool (CPU-based controllers with far higher throughput than hardware OFAs) via Packet-In Edge messages; monitors and separates small DDoS flows (terminated at vSwitches) from large legitimate flows (migrated back to physical paths); overlay automatically phases out when congestion clears; achieves **>70% controller congestion reduction** on 34-node GENI Clos topology without modifying physical switch hardware.

INDUSTRY EXPERIENCE

Technical Product Manager 07/2018 – 01/2019

Xiaomi

- Defined product requirements for a smart IoT appliance via competitive analysis and user research; assessed technical feasibility and delivered architecture proposals to align engineering and business stakeholders.

Product Manager 09/2016 – 06/2018

Midea Group

- Led end-to-end technical PM for **Fun Oven II** — overseeing ML model training & deployment, Docker services on Alibaba Cloud, mobile app integration, and culinary algorithm design; product won the **Red Dot Design Award** and debuted at 2018 AWE Show · Red Dot ↗ · PR Newswire ↗

Software Engineer II 06/2015 – 08/2016

Cisco Systems, Inc.

- Developed NPU/ASIC features for a 100 Gbps core router (Azure, AWS, Alibaba Cloud) using Broadcom BCM SDK.
- Contributed to QoS and ECMP modules, reducing forwarding latency under heavy load.

Software Engineer Intern 10/2014 – 05/2015

Broadcom (Emulex)

- Replaced polling-based data acquisition with DMA + circular-buffer architecture in an RTOS pipeline (**30% CPU load reduction**); ported BladeEngine ASIC management CLI from Linux to Windows Server.

Engineer Intern 01/2014 – 04/2014

Tenova I2S

- Re-architected an RTOS data acquisition pipeline using DMA and circular buffers, **reducing CPU load by 42%** and improving sampling rate and control-loop responsiveness.

PUBLICATIONS

8 papers total, 7 first-author, 5 published at IEEE CCNC, IEEE ICNC, ACM TOIT, and Springer SecureComm; 3 under review (incl. ACM TOPS). Full list: johnny-dai-git.github.io/portfolio